

All proofs are deferred to the appendix. A discussion of additional related works is provided in Appendix A. Metastable dynamics and hitting times are studied in Appendices B-C, optimization dynamics are analyzed in Appendix D, and learning-theoretic lower bounds are given in Appendix E.

2 Metastable Dynamics and Reasoning

2.1 CoT as Markov Chains

Our key insight to understanding inference-time search is to frame CoT reasoning as a metastable Markov process over an underlying linguistic model. Each state represents a logical assertion (e.g., a sentence or mathematical expression rather than a single token), and state transitions correspond to reasoning steps. The model distinguishes between **easy/trivial reasoning steps**, which form dense local clusters of roughly equivalent meaning, and **hard reasoning steps**, which form sparse connections between clusters of small probability $O(\varepsilon)$. Reasoning paths sampled from this process typically spend a long time in each cluster before making a nontrivial jump to another cluster. This leads to a dynamical separation between fast and slow timescales, which we quantitatively study by tuning the degree ε of perturbation.

The setup is formalized as follows. Let $X^\varepsilon = (X_t^\varepsilon)_{t \geq 0}$ be a perturbed family of discrete-time stationary Markov chains on a (large but finite) state space S with transition kernel p^ε , such that p^ε uniformly converges to p^0 as $\varepsilon \rightarrow 0$. We assume X^ε is recurrent for all $\varepsilon \geq 0$ and irreducible for all $\varepsilon > 0$; also, X^0 is reducible and decomposes S into K disjoint p^0 -ergodic components $C_1, \dots, C_K$. We set $M := \max_k |C_k|$ and assume that $\min_k |C_k| = \Theta(M)$ and $K \leq \text{poly}(M)$. Moreover, we denote the stochastic complement of p^ε corresponding to C_k by the matrix $\mathbf{S}_{kk}^\varepsilon$; see Appendix B for definitions. The stationary distributions of p^ε , $\mathbf{S}_{kk}^\varepsilon$ are denoted by $\pi^\varepsilon, \pi_k^\varepsilon$ and we set $\mu_k := \pi_k^0$.

Assumption 1 (dense clusters). *For each $\mathbf{S}_{kk}^\varepsilon$, the pseudo-spectral gap $\gamma^\dagger(\mathbf{S}_{kk}^\varepsilon) \geq \gamma > 0$ and the stationary measure π_k^ε satisfies $\pi_k^\varepsilon(x) = \Theta(1/M)$ for all $x \in C_k$.*

We give verifiable conditions on the unperturbed kernel p^0 which guarantee Assumption 1 in Proposition B.8.

We further denote $E_0 = \text{supp } p^0$ and assume that $E = \text{supp } p^\varepsilon$ is fixed for all $\varepsilon > 0$. A reasoning path $X_{0:T}$ is termed *valid* if $(X_{t-1}, X_t) \in E$ for all $t \in [T]$. The set of sparse edges is denoted by $E_s = E \setminus E_0$.

Assumption 2 (sparse edges). *There are at most d_{out} sparse edges from each of at most n_{out} sources in C_k , and there is at most one sparse edge between any two distinct clusters, with at least one sparse edge from each cluster. Moreover, $p^\varepsilon(y|x) \propto \varepsilon$ for each $(x, y) \in E_s$ with proportionality constant bounded above and below w.r.t. M, K , and $p^\varepsilon(z|x)$ for $(x, z) \in E_0$ all decrease proportionally with ε .*

2.2 Reasoning Task

The reasoner is given a pair of input and output states $(X_{\text{in}}, X_{\text{out}})$ sampled from a distribution $\mathcal{D}$ on $S \times S$. The goal of the reasoner is to find a valid path from X_{in} to X_{out} . We are thus interested in the hitting time of CoT generation to understand inference-time computation. The overall difficulty of the task is measured by the minimum number of hard reasoning steps needed to reach X_{out} from X_{in} ; longer reasoning chains will require more sparse transitions. We assume the average difficulty of the task is lower bounded:

Assumption 3. For $(X_{\text{in}}, X_{\text{out}}) \sim \mathcal{D}$ and any valid path $X_{0:T}$ with $X_0 = X_{\text{in}}$ and $X_T = X_{\text{out}}$, it holds that

$$\mathbb{E}_{\mathcal{D}}[\min |X_{0:T} \cap E_s|] = \Omega(K).$$

Note that X_{out} is already known in our setting. For example, for theorem proving, X_{in} is the problem statement and X_{out} is the QED symbol; or when asked a “why” question, X_{out} could be the conclusion, “That is why...” Nonetheless, for many reasoning problems the answer is unknown and must be deduced or computed. We incorporate this aspect by introducing a ‘logical computation’ task in Section 5.

2.3 Metastable Dynamics

The *hitting time* and *return time* of X^ε to a set $A \subseteq S$ are defined as $\tau_A^\varepsilon = \inf\{t \geq 0 : X_t^\varepsilon \in A\}$, $\bar{\tau}_A^\varepsilon = \inf\{t > 0 : X_t^\varepsilon \in A\}$, respectively. Probabilities and expectations conditioned on the initial state x are denoted as $\mathbb{P}_x, \mathbb{E}_x$, etc.

In the context of perturbed Markov chains, a subset $M \subset S$ is defined as a *metastable system* (Bovier et al., 2002) if

$$\lim_{\varepsilon \rightarrow 0} \sup_{x \in M, y \notin M} \frac{\mathbb{P}_x(\bar{\tau}_{M \setminus \{x\}}^\varepsilon < \bar{\tau}_x^\varepsilon)}{\mathbb{P}_y(\bar{\tau}_M^\varepsilon < \bar{\tau}_y^\varepsilon)} = 0. \quad (1)$$

That is, it is much easier to return to M than to transition between different states in M . The following result, obtained from our perturbative analysis in Appendices B-C, will motivate the distillation scheme described in Section 4.

Proposition 2.1. Any subset $S_\circ = \{x_1, \dots, x_K\} \subset S$ of cluster representatives $x_k \in C_k$ constitutes a metastable system for X^ε in the sense of (1) as $M \rightarrow \infty$.

Meta-chain. The coarse-grained dynamics of X^ε over long timescales is captured by its effective **metastable representation** $X_\star^\varepsilon$ (Wicks and Greenwald, 2005; Betz and Le Roux, 2016), which acts as a compression of the full chain by only retaining information on inter-cluster dynamics. This ‘meta-chain’ is defined on the set of clusters $S_\star = \{C_1, \dots, C_K\}$ with transition kernel

$$q_\star^\varepsilon(C_\ell | C_k) = \sum_{x \in C_k} \mu_k(x)^2 \mathbb{P}_x(\bar{\tau}_{C_\ell}^\varepsilon < \bar{\tau}_x^\varepsilon), \quad k \neq \ell, \quad (2)$$

Algorithm 1 Two-stage Pretraining

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1: set  $\mathbf{W}^{(0)} = \mathbf{0}$ ,  $\eta = O(KM)$ ,  
2:  $T_1 = \tilde{O}(KM^2\varepsilon^{-2})$ ,  $T_2 = \tilde{O}(KM\varepsilon^{-2})$   
3: for  $t = 1, \dots, T_1$  do  
4:    $\mathbf{W}^{(t)} = \mathbf{W}^{(t-1)} + \eta \nabla \mathbb{E}_{X_0, X_1} [\log \hat{p}_{\mathbf{W}^{(t-1)}}(X_1|X_0)]$   
5: end for  
6:  $w_{ij}^{(T_1)} \leftarrow -\infty$  if  $\hat{p}_{ij}^{(T_1)} < c_{\text{thres}}\varepsilon$  (thresholding)  
7: for  $t = T_1 + 1, \dots, T_2$  do  
8:    $\mathbf{W}^{(t)} = \mathbf{W}^{(t-1)} + \eta \nabla \mathbb{E}_{X_0, X_1} [\log \hat{p}_{\mathbf{W}^{(t-1)}}(X_1|X_0)]$   
9: end for
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and $q_\star^\varepsilon(C_k|C_k)$ such that the conditional probabilities sum to 1. We emphasize that $X_\star^\varepsilon$ faithfully characterizes cluster escape probabilities (see Proposition 4.1) but is *not* a one-to-one copy of the cluster transitions of X^ε , which generally cannot be uniquely defined as a Markov chain. For example, $q_\star^\varepsilon$ is asymptotically reversible and always positive regardless of the actual arrangement of sparse edges (Proposition C.8). To provide further intuition, we state and discuss the following assumption.

Assumption 4 (uniform escape of $X_\star^\varepsilon$). *For all $k \neq \ell$,*

$$q_\star^\varepsilon(C_\ell|C_k) = \Omega(\varepsilon/M). \quad (3)$$

Equation (3) holds if there exists a sparse edge from C_k to C_ℓ (Corollary C.11), but it may well hold even if C_k, C_ℓ are not directly connected. For example, if the sparse edges are arranged as a cycle on $S_\star$, escaping C_k implies that all other clusters C_ℓ will be hit before the process returns to C_k , and so Assumption 4 is satisfied. Hence Assumption 4 naturally guarantees that it is easy to explore the entire state space from any starting cluster.¹

3 Search Improves the Pretrained Model

3.1 Pretraining the Base (World) Model

We equate pretraining the base model with learning the underlying transition kernel p^ε . Indeed, if the context window of an LLM is restricted to the tokens in the previous state, next-token prediction recursively defines a distribution over the following state, and further over reasoning chains of arbitrary length. We encode each state $x \in S$ as a one-hot vector in $\mathbb{R}^{|S|}$ also denoted by x and write $p_{ij}^\varepsilon = p^\varepsilon(e_j|e_i)$. For the model, we consider a simple linear softmax predictor:

$$\hat{p}_{\mathbf{W}}(\cdot|x) = \text{softmax}(\langle \mathbf{W}, x \rangle), \quad \mathbf{W} \in \mathbb{R}^{|S| \times |S|}.$$

¹On the other hand, if the meta-chain has poorly connected regions, then $X_\star^\varepsilon$ itself is amenable to metastability analysis, leading to a hierarchy of metastable representations at increasingly faster timescales (Wicks and Greenwald, 2005).